

Low-Conductor Suppression in Centered Sieve Kernels: Power-Short Rough Boxes and a Shifted-Divisor Quadratic Design

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Abstract

Let Q be the odd squarefree product of the primes in a finite window and let $G = (\mathbb{Z}/Q\mathbb{Z})^\times$. We study the centered local pair kernel obtained by excluding $ab + h \equiv 0 \pmod{p}$ for every $p \mid Q$. A preceding finite Fourier calculation gives exact multipliers indexed by Dirichlet characters and weighted by their conductors. Here we combine that kernel structure with the classical reduction of an imprimitive character to its primitive conductor and with Pólya–Vinogradov and Burgess bounds.

For the quadratic window $Q = \prod_{w < p \leq w^2} p$ and arbitrary integer intervals I, J of common length $L = \lceil Q^{1-\eta} \rceil$, we prove unconditionally, as $w \rightarrow \infty$, that the normalized complete fully-active-shift root mean square is $o(1)$ for every fixed $0 < \eta < 1$. The same conclusion holds when averaging the fully active unit shifts contained in any interval of $H = \lfloor Q^{1/\log w} \rfloor$ consecutive integers. For the prescribed local shift $h = 2$, a conductor cutoff combining Burgess at low conductor with the exact kernel multiplier at high conductor gives $\mathcal{B}_2(I, J) = o(L^2)$ for every fixed $0 < \eta < 4/5$. The endpoints are not asserted and the ranges are not claimed to be optimal.

For prime windows with every $p \geq 5$, we also give a finite shifted-divisor Selberg-form square which pointwise majorizes a residue-avoidance indicator and annihilates every one-prime Fourier mode exactly; its nonconstant spectrum is supported only on two-prime conductors. A rational certificate at $Q = 5 \cdot 7 \cdot 11$ verifies the identities conductor by conductor. All asymptotic box theorems use coefficient one on the unit support; the finite designed weight is local and not prime-sensitive. The paper contains no von Mangoldt weight, no prime-sensitive Type II estimate, and no implication toward a twin-prime lower bound.

Keywords: sieve kernel; Dirichlet character; Burgess bound; conductor; rough interval; Selberg-form square; parity boundary.

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1 Introduction and scope

This paper makes one specific transition: from an exact finite Fourier description of a centered Chinese-remainder pair kernel to asymptotic cancellation on genuinely power-short factor boxes. The character-sum input is classical. The new point is its conductor-by-conductor assembly against the exact multipliers and shift Parseval identity of the companion paper [16]. In particular, we do not prove a new Pólya–Vinogradov or Burgess estimate.

Let

$$Q = \prod_{w < p \leq w^2} p, \quad G = (\mathbb{Z}/Q\mathbb{Z})^\times, \quad g = \varphi(Q), \quad \kappa = \prod_{p|Q} \frac{p-2}{p-1}. \quad (1)$$

For integer intervals I, J of the same length $L < Q$, and for $(h, Q) = 1$, write

$$\mathcal{B}_h(I, J) = \sum_{\substack{m \in I, n \in J \\ (mn, Q) = 1}} \left\{ \kappa^{-1} \mathbf{1}_{(mn+h, Q) = 1} - 1 \right\}. \quad (2)$$

Thus the coefficient on the unit support is identically one. Arbitrary bounded coefficients are not included in the theorems below. The centering in (2) is the exact complete-group centering, not an empirical centering chosen from the observed box.

1.1 Main results

Fix $0 < \eta < 1$ and put

$$L = \left\lceil Q^{1-\eta} \right\rceil. \quad (3)$$

Uniformly in the locations of I and J , we prove

$$\frac{1}{L^2} \left(\frac{1}{g} \sum_{\ell \in G} |\mathcal{B}_{2\ell}(I, J)|^2 \right)^{1/2} = o(1) \quad (w \rightarrow \infty). \quad (4)$$

This is a complete fully-active-shift mean: $(\ell, Q) = 1$ ensures that every prime in the window is active for $h = 2\ell$. It is not a conclusion for each shift separately.

The complete average can be sampled on a much shorter interval. Let $\mathcal{L}$ be any interval of

$$H = \left\lfloor Q^{1/\log w} \right\rfloor \quad (5)$$

consecutive integers and set $H_Q = \#\{\ell \in \mathcal{L} : (\ell, Q) = 1\}$. Then, again for every fixed $0 < \eta < 1$ and uniformly in the locations of all three intervals,

$$\frac{1}{L^2} \left(\frac{1}{H_Q} \sum_{\substack{\ell \in \mathcal{L} \\ (\ell, Q) = 1}} |\mathcal{B}_{2\ell}(I, J)|^2 \right)^{1/2} = o(1). \quad (6)$$

Although $H = Q^{o(1)}$, this remains an averaged theorem and does not select a prescribed member of $\mathcal{L}$.

For the prescribed local shift, the result is necessarily stated separately. For every fixed $0 < \eta < 4/5$,

$$\frac{\mathcal{B}_2(I, J)}{L^2} = o(1), \quad (7)$$

uniformly in the locations of I, J . The range comes from choosing a conductor cutoff $R = Q^\alpha$ with

$$\eta < \alpha < 4(1 - \eta) : \quad (8)$$

after choosing such an α , one fixes a sufficiently large Burgess parameter so that the slightly stronger finite-parameter upper inequality also holds. Low conductors are treated by Burgess and high conductors by the exact kernel multiplier. We make no assertion at $\eta = 1$ or $\eta = 4/5$, and neither strict range is claimed to be optimal or to be a barrier for other methods.

There is also a finite algebraic result of a different kind. We construct a nonnegative shifted-divisor square which majorizes the relevant local residue-avoidance indicator and has every nonconstant one-prime Fourier mode equal to zero. Its remaining nonconstant spectrum lies on two-prime conductors. This construction explains both what an ordinary same-window divisor square cannot change on the unit group and what cross-prime coupling can change. It is a finite spectral design, not a prime-detecting sieve asymptotic.

1.2 Where the power saving comes from

For a character $\chi \in \widehat{G}$, let q_χ be the product of the local primes on which χ is nontrivial. Since Q is squarefree, χ is induced from a primitive character of conductor exactly q_χ . The companion calculation gives, for $\chi \neq \mathbf{1}$, an exact kernel coefficient of modulus

$$d(\chi) = \prod_{p|q_\chi} \frac{1}{p-2}, \quad (9)$$

together with an exact diagonal Parseval formula after averaging the fully active shifts [16]. Reduction to the primitive conductor converts each rough interval transform into ordinary incomplete character sums. Pólya–Vinogradov, as presented for example in [12, Section 9.4], gives a useful intermediate power-short range. The final ranges in all three main statements use the Burgess estimate for squarefree, hence cubefree, conductors [2, 3]. At high conductor, (9) itself supplies the decay. The contribution here is this division of labor and the resulting exponents for the particular CRT pair kernel, not any strengthening of those classical character-sum theorems.

The distinction from earlier sifted character sums is worth recording. Schlag–Puchta inserted a sifting device into a character sum in a specific application [14]; Puchta’s short character sums over primes use a large-sieve inequality for prime-supported coefficients together with the Selberg sieve [13]. Our sums instead have constant coefficients on the Q -unit support and a two-variable centered periodic kernel. Those works provide important methodological context, but their prime-supported conclusions are neither inputs to nor consequences of (4)–(7).

Low-conductor exceptional behavior also plays a central role in general Bombieri–Vinogradov theories for multiplicative functions [6, 7]. Those theorems concern distribution in arithmetic progressions, averaged over moduli, under Siegel–Walfisz and related hypotheses. Here the modulus Q is one fixed squarefree product, the averaging in (4) is over local shifts, and the exact conductor weight comes from the kernel. Consequently, the present theorem is not a Bombieri–Vinogradov theorem for a new multiplicative function.

The congruence $mn \equiv -h \pmod{q}$ also places the problem near the literature on modular hyperbolas; see the survey [15]. Our condition, however, is a product of exclusions at the primes dividing Q , followed by exact CRT centering. We use its specialized character factorization and do not claim a new point-count estimate for a modular hyperbola.

Finally, factorability has a precise meaning in results on primes in arithmetic progressions. The large-modulus work of Bombieri et al. [1] and its extensions use structured averages that are far more than a formal factorization of one squarefree integer. Maynard’s fixed-residue-class theorem uses moduli with suitably placed divisors, whereas the well-factorable theorem averages prime-progression errors with suitably well-factorable modulus weights [10, 11]. Neither condition is supplied merely by writing Q as a product of primes. Our shift average is not a well-factorable modulus average, and (7) contains none of the prime-sensitive dispersion or Type II input in those works.

1.3 Selberg form and strict boundary of the claims

The nonnegative shifted-divisor square is motivated by the algebraic form of Selberg sieve weights; standard sieve theory and modern smoothing questions are treated in [5, 8]. Our use is deliberately finite: coefficients are chosen so that specified Fourier modes of a local function vanish exactly. We do not optimize an asymptotic upper- or lower-bound sieve, and the label “Selberg-form” does not turn the weight into a detector of primes.

More generally, every asymptotic box result in this paper concerns the coefficient-one periodic expression (2); the separate quadratic design is a finite local weight. There is no von Mangoldt function Λ , no prime-supported coefficient sequence, and no Type II estimate for such a sequence. An ambient estimate $o(L^2)$ is also not a positive main term on the prime-pair scale. Classical

sieve parity remains a separate obstruction [9], and prime-producing asymptotic sieve arguments require additional information absent here [4]. Therefore none of (4)–(7), nor the finite quadratic design, implies or provides evidence for the twin-prime conjecture. The paper’s claim is narrower: classical incomplete character-sum tools, combined with an exact CRT kernel, now reach power-short unweighted rough boxes and expose several concrete prime-sensitive interfaces that this local model does not control.

2 The centered CRT pair kernel

This section records the finite identities used later. They are included to fix normalizations and to separate the new power-short consequences from the exact complete-period calculation.

2.1 Group, kernel, and conductor

Let $\mathcal{P}$ be a nonempty finite set of odd primes and put

$$Q = \prod_{p \in \mathcal{P}} p, \quad G = (\mathbb{Z}/Q\mathbb{Z})^\times, \quad g = |G| = \varphi(Q), \quad r = \omega(Q). \quad (10)$$

We use normalized Haar expectation

$$\mathbb{E}_G f = \frac{1}{g} \sum_{a \in G} f(a).$$

By the Chinese remainder theorem,

$$G \cong \prod_{p|Q} \mathbb{F}_p^\times, \quad \widehat{G} \cong \prod_{p|Q} \widehat{\mathbb{F}_p^\times}.$$

For $\chi = \bigotimes_{p|Q} \chi_p \in \widehat{G}$, define

$$\text{supp}_{\mathcal{P}}(\chi) = \{p \mid Q : \chi_p \neq \mathbf{1}\}, \quad q_\chi = \prod_{p \in \text{supp}_{\mathcal{P}}(\chi)} p. \quad (11)$$

We extend χ by zero off the units. Because Q is squarefree and every nontrivial character modulo a prime is primitive, this Dirichlet character modulo Q is induced by a primitive character χ^* of conductor exactly q_χ . Thus q_χ is both the local-prime-support product and the usual conductor; we use the latter term below.

Let h be an integer with $(h, Q) = 1$. Set

$$K_h(a, b) = \prod_{p|Q} \mathbf{1}_{a_p b_p \neq -h \bmod p}, \quad (12)$$

$$\kappa = \prod_{p|Q} \frac{p-2}{p-1}, \quad (13)$$

$$K_h^\circ(a, b) = \kappa^{-1} K_h(a, b) - 1. \quad (14)$$

The normalization is exact: for either variable fixed,

$$\mathbb{E}_{a \in G} K_h^\circ(a, b) = \mathbb{E}_{b \in G} K_h^\circ(a, b) = 0. \quad (15)$$

For a nontrivial character define

$$d(\chi) = d(q_\chi) := \prod_{p|q_\chi} \frac{1}{p-2}, \quad \Xi = \prod_{p|Q} \frac{p}{p-2}. \quad (16)$$

We use the empty-product convention $d(1) = 1$.

Proposition 2.1 (Exact conductor expansion). *For $(h, Q) = 1$ and $a, b \in G$,*

$$K_h^\circ(a, b) = \sum_{\substack{\chi \in \widehat{G} \\ \chi \neq \mathbf{1}}} c_h(\chi) \chi(a) \chi(b), \quad (17)$$

where

$$c_h(\chi) = (-1)^{\omega(q_\chi)} d(\chi) \overline{\chi(-h)}. \quad (18)$$

For every nonempty $q \mid Q$,

$$\#\{\chi \in \widehat{G} : q_\chi = q\} = \prod_{p \mid q} (p-2), \quad (19)$$

$$\sum_{q_\chi = q} |c_h(\chi)| = 1, \quad (20)$$

$$\sum_{q_\chi = q} |c_h(\chi)|^2 = d(q). \quad (21)$$

Proof. For $p \nmid h$ and $t \in \mathbb{F}_p^\times$, character orthogonality gives

$$\frac{p-1}{p-2} \mathbf{1}_{t \neq -h} = 1 - \frac{1}{p-2} \sum_{\substack{\chi_p \in \widehat{\mathbb{F}_p^\times} \\ \chi_p \neq \mathbf{1}}} \overline{\chi_p(-h)} \chi_p(t).$$

Multiplying over $p \mid Q$ and removing the all-trivial term proves (17). At each $p \mid q$ there are $p-2$ nontrivial local characters, each with coefficient modulus $(p-2)^{-1}$. Multiplication over the support proves (19)–(21). $\square$

2.2 Interval transforms and shift Parseval

Let $I, J \subset \mathbb{Z}$ be intervals of a common length $L < Q$. Define

$$A_I(\chi) = \sum_{\substack{m \in I \\ (m, Q) = 1}} \chi(m), \quad A_J(\chi) = \sum_{\substack{n \in J \\ (n, Q) = 1}} \chi(n), \quad (22)$$

$$U_I = \#\{m \in I : (m, Q) = 1\}, \quad U_J = \#\{n \in J : (n, Q) = 1\}. \quad (23)$$

The fixed-shift box form is

$$\mathcal{B}_h(I, J) = \sum_{\substack{m \in I, n \in J \\ (mn, Q) = 1}} K_h^\circ(m, n). \quad (24)$$

Substitution of (17) gives

$$\mathcal{B}_h(I, J) = \sum_{\chi \neq \mathbf{1}} c_h(\chi) A_I(\chi) A_J(\chi). \quad (25)$$

For $\ell \in G$, every prime dividing Q is active for $h = 2\ell$. The phase in (18) is a character of ℓ , so orthogonality yields the following identity.

Proposition 2.2 (Complete fully-active-shift Parseval). *For intervals I, J as above,*

$$\frac{1}{g} \sum_{\ell \in G} |\mathcal{B}_{2\ell}(I, J)|^2 = \sum_{\chi \neq \mathbf{1}} d(\chi)^2 |A_I(\chi)|^2 |A_J(\chi)|^2. \quad (26)$$

Equivalently, the contribution of exact conductor q is

$$\mathcal{E}_q(I, J) = d(q)^2 \sum_{q_\chi = q} |A_I(\chi)|^2 |A_J(\chi)|^2 \geq 0. \quad (27)$$

Proof. By (18),

$$c_{2\ell}(\chi) = (-1)^{\omega(q_\chi)} d(\chi) \overline{\chi(-2)} \overline{\chi(\ell)}.$$

After expanding the square in (25), the average of $\overline{\chi(\ell)} \chi(\ell)$ over G is $\mathbf{1}_{\chi=\psi}$. $\square$

2.3 The high-conductor side

The exact multiplier already controls high conductors. Indeed,

$$d(\chi) = \frac{1}{q_\chi} \prod_{p|q_\chi} \frac{p}{p-2} \leq \frac{\Xi}{q_\chi}. \quad (28)$$

Proposition 2.3 (High-conductor estimates). *For $R \geq 1$, let the superscript $> R$ denote the portion of (25) or (26) with $q_\chi > R$. Then*

$$|\mathcal{B}_h^{>R}(I, J)| \leq \frac{\Xi g}{R} \sqrt{U_I U_J} \leq \frac{\Xi g L}{R}, \quad (29)$$

$$\mathcal{E}^{>R}(I, J) \leq \frac{\Xi^2 g L^3}{R^2}, \quad (30)$$

where $\mathcal{E}^{>R}$ denotes the corresponding part of the normalized complete-shift mean square.

Proof. Parseval on G gives

$$\sum_{\chi \in \widehat{G}} |A_I(\chi)|^2 = g U_I, \quad \sum_{\chi \in \widehat{G}} |A_J(\chi)|^2 = g U_J. \quad (31)$$

Use (28), Cauchy–Schwarz, and (31) in (25) to obtain (29). For (30), use $|A_J(\chi)| \leq L$ and sum $|A_I(\chi)|^2$ by (31). $\square$

2.4 Sampling a short shift interval

Let $\mathcal{L}$ be an interval of H consecutive integers and put

$$H_Q = \#\{\ell \in \mathcal{L} : (\ell, Q) = 1\}, \quad \delta_Q = \frac{g}{Q}. \quad (32)$$

Direct inclusion–exclusion gives

$$|H_Q - H\delta_Q| \leq 2^r. \quad (33)$$

Set

$$E(Q) = \kappa^{-2} 4^r + 2\kappa^{-1} 3^r + 2^r. \quad (34)$$

Proposition 2.4 (Deterministic shift completion). *For every $I, J, \mathcal{L}$ as above,*

$$\left| \sum_{\substack{\ell \in \mathcal{L} \\ (\ell, Q)=1}} |\mathcal{B}_{2\ell}(I, J)|^2 - \frac{H}{Q} \sum_{\ell \in G} |\mathcal{B}_{2\ell}(I, J)|^2 \right| \leq E(Q) U_I^2 U_J^2. \quad (35)$$

If $H\delta_Q \geq 2^{r+1}$, then

$$\frac{1}{L^2} \left(\frac{1}{H_Q} \sum_{\substack{\ell \in \mathcal{L} \\ (\ell, Q)=1}} |\mathcal{B}_{2\ell}(I, J)|^2 \right)^{1/2} \ll \frac{\sqrt{\mathcal{E}(I, J)}}{L^2} + \sqrt{\frac{E(Q)}{H\delta_Q}}, \quad (36)$$

where $\mathcal{E}(I, J)$ is the left side of (26).

Proof. For a fixed unit product $t = mn \bmod Q$, the binary condition in shift 2ℓ excludes the residue $-t/2$ modulo every $p \mid Q$. After expanding $|\mathcal{B}_{2\ell}|^2$, a two-kernel term together with $(\ell, Q) = 1$ excludes at most three residues per prime; a one-kernel term excludes at most two; and the unit condition excludes one. Inclusion–exclusion therefore gives discrepancies bounded by 4^r , 3^r , and 2^r , respectively. Their coefficients are κ^{-2} , $-\kappa^{-1}$, $-\kappa^{-1}$, and 1, which proves (35). Under the stated hypothesis, (33) gives $H_Q \geq H\delta_Q/2$. Divide (35) by H_Q , use $U_I U_J \leq L^2$, and take square roots. $\square$

2.5 The quadratic prime window

All logarithms in the asymptotic statements are natural. The asymptotic results use

$$\mathcal{P}(w) = \{p : w < p \leq w^2\}, \quad Q = \prod_{p \in \mathcal{P}(w)} p. \quad (37)$$

The prime number theorem and Mertens' theorem imply

$$\log Q = (1 + o(1))w^2, \quad (38)$$

$$r = (1 + o(1)) \frac{w^2}{2 \log w}, \quad (39)$$

$$\delta_Q = \frac{1}{2} + o(1), \quad \kappa = \frac{1}{2} + o(1), \quad \Xi = 4 + o(1). \quad (40)$$

In particular, for every fixed $c > 1$,

$$c^r = Q^{o(1)}. \quad (41)$$

For the shift length

$$H = \left\lfloor Q^{1/\log w} \right\rfloor, \quad (42)$$

one has

$$\frac{E(Q)}{H\delta_Q} = \exp \left(-(1 - \log 2 + o(1)) \frac{w^2}{\log w} \right) = o(1). \quad (43)$$

Thus the completion interval is $Q^{o(1)}$, yet it dominates the explicit completion cost.

3 Primitive reduction and power-short rough boxes

The exact kernel identities become useful on power-short intervals only after one controls the interval transforms at low conductor. This section supplies that input by a standard imprimitive-to-primitive reduction followed by classical character-sum estimates. Neither the Pólya–Vinogradov inequality nor the Burgess bound used below is new; our only use of them is through the rough transform dictated by the kernel.

3.1 The rough interval transform

Let $q > 1$ divide the squarefree modulus Q , and let χ^* be a primitive Dirichlet character of conductor q . Inflating χ^* to modulus Q gives the character

$$\chi_Q(n) = \chi^*(n) \mathbf{1}_{(n, Q/q)=1}. \quad (44)$$

Thus χ_Q vanishes off the units modulo Q and, on G , has exact local support product q . For an integer interval I , define its *Q -rough character transform* by

$$\mathfrak{A}_I(Q; \chi^*) := \sum_{n \in I} \chi_Q(n) = \sum_{\substack{n \in I \\ (n, Q)=1}} \chi^*(n). \quad (45)$$

Under the identification in (11), this is exactly $A_I(\chi)$ from (22).

For $d \mid Q/q$, put

$$I/d = \{m \in \mathbb{Z} : dm \in I\}, \quad L_d = |I/d|. \quad (46)$$

The set I/d is an integer interval, possibly empty, and if $L = |I|$ then $L_d \leq L/d + 1$. Möbius inversion in (44) gives the following exact reduction.

Lemma 3.1 (Imprimitive-to-primitive reduction). *For every $q \mid Q$, $q > 1$, every primitive $\chi^* \pmod{q}$, and every integer interval I ,*

$$\mathfrak{A}_I(Q; \chi^*) = \sum_{d \mid Q/q} \mu(d) \chi^*(d) \sum_{m \in I/d} \chi^*(m). \quad (47)$$

Proof. Since $(d, q) = 1$ for $d \mid Q/q$, inclusion–exclusion and complete multiplicativity give

$$\begin{aligned} \mathfrak{A}_I(Q; \chi^*) &= \sum_{n \in I} \chi^*(n) \sum_{d \mid (n, Q/q)} \mu(d) \\ &= \sum_{d \mid Q/q} \mu(d) \sum_{\substack{n \in I \\ d \mid n}} \chi^*(n) = \sum_{d \mid Q/q} \mu(d) \chi^*(d) \sum_{m \in I/d} \chi^*(m). \end{aligned}$$

This also covers terms for which $(m, q) > 1$, because both sides of $\chi^*(dm) = \chi^*(d)\chi^*(m)$ then vanish. $\square$

We record three consequences in increasing analytic strength. The first is only periodicity and the triangle inequality. The second and third are the classical Pólya–Vinogradov and Burgess bounds; see, for example, Montgomery and Vaughan [12] and the original work of Burgess [2, 3].

Proposition 3.2 (Three classical bounds for the rough transform). *With the notation above, one has the elementary estimate*

$$|\mathfrak{A}_I(Q; \chi^*)| \leq \sum_{d \mid Q/q} \min\{L_d, q\} \leq \tau(Q/q) \min\{L + 1, q\}, \quad (48)$$

and the Pólya–Vinogradov estimate

$$|\mathfrak{A}_I(Q; \chi^*)| \ll \tau(Q/q) \sqrt{q} \log(2q). \quad (49)$$

For every integer $k \geq 2$ and every $\varepsilon > 0$, the Burgess estimate is

$$\begin{aligned} |\mathfrak{A}_I(Q; \chi^*)| &\ll_{k, \varepsilon} q^{(k+1)/(4k^2)+\varepsilon} \sum_{\substack{d \mid Q/q \\ L_d > 0}} L_d^{1-1/k} \\ &\leq \tau(Q/q) (L + 1)^{1-1/k} q^{(k+1)/(4k^2)+\varepsilon}. \end{aligned} \quad (50)$$

All bounds are uniform in the location of I . The implied constant in (49) is absolute, and that in (50) depends only on k and ε .

Proof. A nonprincipal character sums to zero over a complete period modulo q . After deleting complete periods, the trivial estimate for an interval of L_d integers is therefore $\min\{L_d, q\}$. Substitution in (47) proves (48).

Pólya–Vinogradov gives

$$\left| \sum_{m \in I/d} \chi^*(m) \right| \ll \sqrt{q} \log(2q)$$

uniformly in I/d . Summing over the $\tau(Q/q)$ divisors proves (49).

Finally, q is squarefree and hence cubefree. The classical cubefree-modulus form of Burgess is consequently available for every fixed integer $k \geq 2$: after complete periods have been removed,

$$\left| \sum_{m \in I/d} \chi^*(m) \right| \ll_{k, \varepsilon} L_d^{1-1/k} q^{(k+1)/(4k^2)+\varepsilon}.$$

Summing this inequality in (47), and then using $L_d \leq L + 1$, proves (50). These steps do not assert a new character-sum estimate. $\square$

Remark 3.3 (Why unweighted intervals matter). The exact reduction (47) uses coefficient one on the rough integers. It does not give the same bounds after inserting arbitrary coefficients, a von Mangoldt weight, or a prime-supported sequence. Such replacements require separate character-sum input.

3.2 Exact aggregation by local support

For $q \mid Q$, put

$$D(q) = \prod_{p \mid q} (p-2) = d(q)^{-1}, \quad \mathfrak{X}(q) = \{\chi \in \widehat{G} : q_\chi = q\}. \quad (51)$$

The characters in $\mathfrak{X}(q)$ are precisely the inflations of primitive characters of conductor q , and $|\mathfrak{X}(q)| = D(q)$. Define the uniform rough-transform bounds

$$\mathcal{U}_I(q) = \max_{\chi \in \mathfrak{X}(q)} |A_I(\chi)|, \quad \mathcal{U}_J(q) = \max_{\chi \in \mathfrak{X}(q)} |A_J(\chi)|. \quad (52)$$

The following bookkeeping is a direct support-by-support consequence of the exact kernel expansion recalled from Wang [16]. It is stated explicitly because fixed-shift and shift-mean arguments use different coefficient masses.

Proposition 3.4 (Support-by-support kernel bounds). *Let $(h, Q) = 1$, and let $\mathcal{B}_{h,q}$ be the portion of (25) with $q_\chi = q$. Then*

$$|\mathcal{B}_{h,q}(I, J)| \leq \mathcal{U}_I(q) \mathcal{U}_J(q). \quad (53)$$

The exact-conductor contribution to the normalized complete fully-active-shift mean square satisfies

$$\mathcal{E}_q(I, J) \leq d(q) \mathcal{U}_I(q)^2 \mathcal{U}_J(q)^2. \quad (54)$$

Consequently, for every cutoff $R \geq 1$,

$$|\mathcal{B}_h^{\leq R}(I, J)| \leq \sum_{\substack{q \mid Q \\ 1 < q \leq R}} \mathcal{U}_I(q) \mathcal{U}_J(q), \quad (55)$$

$$\mathcal{E}^{\leq R}(I, J) \leq \sum_{\substack{q \mid Q \\ 1 < q \leq R}} d(q) \mathcal{U}_I(q)^2 \mathcal{U}_J(q)^2. \quad (56)$$

Proof. For exact conductor q , every coefficient in (18) has modulus $d(q)$ and there are $D(q) = d(q)^{-1}$ such characters. The triangle inequality therefore gives (53). Likewise, (27) gives

$$\mathcal{E}_q(I, J) \leq d(q)^2 D(q) \mathcal{U}_I(q)^2 \mathcal{U}_J(q)^2,$$

which is (54). Summation over q proves the remaining assertions. $\square$

Corollary 3.5 (Elementary finite low-conductor envelopes). *Let $s = \omega(q_\chi)$. For every nontrivial $\chi \in \widehat{G}$ and every integer interval I ,*

$$|A_I(\chi)| \leq \varphi(q_\chi) 2^{r-s}. \quad (57)$$

Consequently, for every $R \geq 1$, equal-length intervals I, J , and $(h, Q) = 1$,

$$|\mathcal{B}_h^{\leq R}(I, J)| \leq 5^r R^2, \quad (58)$$

$$\mathcal{E}^{\leq R}(I, J) \leq 17^r R^3. \quad (59)$$

These bounds are deliberately elementary; they are the finite envelopes checked by the accompanying exact certificate, not the estimates used for the final Burgess ranges.

Proof. In each term of (47), delete complete periods modulo q_χ . The remaining interval has length below q_χ and contains at most $\varphi(q_\chi)$ unit residues. There are 2^{r-s} divisors of Q/q_χ , which proves (57).

Use (55) and (56). Since

$$\sum_{q|Q} 4^{-\omega(q)} = \left(\frac{5}{4}\right)^r,$$

the fixed-shift sum is at most

$$R^2 4^r \sum_{q|Q} 4^{-\omega(q)} = 5^r R^2.$$

For the energy, the primewise inequality

$$\frac{(p-1)^4}{p-2} \leq p^3$$

gives $d(q)\varphi(q)^4 \leq q^3$. Therefore

$$\sum_{\substack{q|Q \\ q \leq R}} d(q)\varphi(q)^4 16^{r-\omega(q)} \leq R^3 16^r \sum_{q|Q} 16^{-\omega(q)} = 17^r R^3.$$

□

3.3 Low-conductor suppression and a conductor-tail estimate

We now specialize to the quadratic window (37) and let $w \rightarrow \infty$. The next lemma isolates the two estimates needed in both applications.

Lemma 3.6 (Burgess suppression below a power cutoff). *Let $\lambda > 0$ and $\alpha > 0$ be fixed with*

$$\alpha < 4\lambda. \tag{60}$$

There is a constant $c = c(\lambda, \alpha) > 0$ such that, uniformly for every integer interval I of length $L \geq Q^\lambda$ and every $\chi \in \widehat{G}$ with $1 < q_\chi \leq Q^\alpha$,

$$\frac{|A_I(\chi)|}{L} \leq Q^{-c+o(1)}. \tag{61}$$

The $o(1)$ is uniform in I and χ .

Proof. Choose an integer $k \geq 2$ so large that

$$\frac{\alpha(k+1)}{4k} < \lambda. \tag{62}$$

Then choose $\varepsilon > 0$ small enough that

$$c_0 := \frac{\lambda}{k} - \alpha \left(\frac{k+1}{4k^2} + \varepsilon \right) > 0.$$

By (50), $L \geq Q^\lambda$, and $q_\chi \leq Q^\alpha$,

$$\begin{aligned} \frac{|A_I(\chi)|}{L} &\ll_{k,\varepsilon} \tau(Q/q_\chi) L^{-1/k} q_\chi^{(k+1)/(4k^2)+\varepsilon} \\ &\leq \tau(Q) Q^{-c_0}. \end{aligned}$$

Here replacing $L+1$ by L changes only an absolute factor. Finally, (41) with $c = 2$ gives $\tau(Q) = 2^{\omega(Q)} = Q^{o(1)}$. Taking, for example, any fixed $c < c_0$ proves the claim for all sufficiently large w . □

The complete shift mean square has a stronger high-conductor estimate than the pointwise operator cutoff because its exact-conductor terms are nonnegative and carry the mass $d(q)$.

Lemma 3.7 (Tail of the exact-conductor mass). *For every fixed $\alpha > 0$,*

$$\sum_{q|Q} d(q) = \kappa^{-1} = 2 + o(1), \quad (63)$$

$$\sum_{\substack{q|Q \\ q > Q^\alpha}} d(q) \ll Q^{-\alpha/\log w} = \exp\left(-(\alpha + o(1))\frac{w^2}{\log w}\right). \quad (64)$$

Proof. The first identity follows from

$$\sum_{q|Q} d(q) = \prod_{p|Q} \left(1 + \frac{1}{p-2}\right) = \prod_{p|Q} \frac{p-1}{p-2} = \kappa^{-1},$$

and (40).

For the tail, take $\theta = 1/\log w$. Rankin's trick gives

$$\begin{aligned} \sum_{\substack{q|Q \\ q > Q^\alpha}} d(q) &\leq Q^{-\alpha\theta} \sum_{q|Q} q^\theta d(q) \\ &= Q^{-\alpha/\log w} \prod_{p|Q} \left(1 + \frac{p^{1/\log w}}{p-2}\right). \end{aligned}$$

For $w < p \leq w^2$, one has $p^{1/\log w} \leq e^2$. Mertens' theorem gives $\sum_{w < p \leq w^2} p^{-1} = \log 2 + o(1)$, so the last product is $O(1)$. Equation (38) proves the final expression in (64). $\square$

3.4 All power lengths after complete or short shift averaging

Theorem 3.8 (Complete-shift RMS on every fixed power length). *Fix $0 < \eta < 1$. Let I, J be arbitrary integer intervals of a common length L satisfying*

$$Q^{1-\eta} \leq L < Q. \quad (65)$$

Then, uniformly in the locations of I and J ,

$$\frac{1}{L^2} \left(\frac{1}{g} \sum_{\ell \in G} |\mathcal{B}_{2\ell}(I, J)|^2 \right)^{1/2} = o(1). \quad (66)$$

Thus the complete fully-active-shift RMS reaches every fixed exponent $1 - \eta > 0$. No endpoint assertion is made at $\eta = 1$.

Proof. Put $\lambda = 1 - \eta$ and choose a fixed

$$0 < \alpha < \min\{1, 4\lambda\}. \quad (67)$$

Let $R = Q^\alpha$. By lemma 3.6, the normalized transform of every character with $1 < q_\chi \leq R$ is at most $Q^{-c+o(1)}$ for some $c > 0$. Using (56) and (63),

$$\frac{\mathcal{E}^{\leq R}(I, J)}{L^4} \leq Q^{-4c+o(1)} \sum_{q|Q} d(q) = Q^{-4c+o(1)}. \quad (68)$$

For $q > R$, the trivial bounds $|A_I(\chi)|, |A_J(\chi)| \leq L$ and the exact number $D(q) = d(q)^{-1}$ of characters give

$$\frac{\mathcal{E}^{>R}(I, J)}{L^4} \leq \sum_{\substack{q|Q \\ q>R}} d(q)^2 D(q) = \sum_{\substack{q|Q \\ q>Q^\alpha}} d(q) = o(1) \quad (69)$$

by lemma 3.7. Adding (68) and (69), and using the exact Parseval identity (26), proves (66). $\square$

The deterministic completion from Wang [16] preserves the preceding conclusion while using only a subpower number of shift parameters.

Corollary 3.9 (A subpower shift interval). *Under the hypotheses of theorem 3.8, let $\mathcal{L}$ be any interval of*

$$H = \lfloor Q^{1/\log w} \rfloor \quad (70)$$

consecutive integers, and let $H_Q = \#\{\ell \in \mathcal{L} : (\ell, Q) = 1\}$. Then

$$\frac{1}{L^2} \left(\frac{1}{H_Q} \sum_{\substack{\ell \in \mathcal{L} \\ (\ell, Q)=1}} |\mathcal{B}_{2\ell}(I, J)|^2 \right)^{1/2} = o(1), \quad (71)$$

uniformly in the locations of all three intervals. Moreover, $H = Q^{o(1)}$ and $H/Q \rightarrow 0$.

Proof. Equations (39) and (42) show that $H\delta_Q \geq 2^{r+1}$ for all sufficiently large w . Hence (36) bounds the left side of (71) by an absolute constant times the left side of (66), plus

$$\sqrt{\frac{E(Q)}{H\delta_Q}}.$$

The first term tends to zero by theorem 3.8, and the second does so by (43). Finally, $1/\log w \rightarrow 0$ while $Q \rightarrow \infty$, proving the assertions about H . $\square$

3.5 The fixed shift and the 4/5 cutoff range

The shift mean square can discard a high-conductor tail through its small total $d(q)$ -mass. At one prescribed shift, phases need not cancel, so we instead use the exact high-conductor operator multiplier (28). The resulting compatibility condition explains the exponent below.

Theorem 3.10 (Fixed local shift on the Burgess cutoff range). *Fix $0 < \eta < 4/5$, and let I, J be arbitrary integer intervals of a common length L satisfying (65). Then*

$$\frac{|\mathcal{B}_2(I, J)|}{L^2} = o(1) \quad (72)$$

uniformly in the locations of I and J .

Proof. Again put $\lambda = 1 - \eta$. The inequality $\eta < 4/5$ is equivalent to $\eta < 4\lambda$. We may therefore choose a fixed exponent α such that

$$\eta < \alpha < \min\{1, 4\lambda\}, \quad R = Q^\alpha. \quad (73)$$

For the low-conductor part, lemma 3.6 and (55) give

$$\frac{|\mathcal{B}_2^{\leq R}(I, J)|}{L^2} \leq \sum_{\substack{q|Q \\ 1 < q \leq R}} \frac{\mathcal{U}_I(q)}{L} \frac{\mathcal{U}_J(q)}{L}$$

$$\leq \tau(Q)Q^{-2c+o(1)} = o(1), \quad (74)$$

where (41) absorbs the number of divisors of Q .

For the high-conductor part, (29), $U_I, U_J \leq L$, $g \leq Q$, and $\Xi = O(1)$ give

$$\frac{|\mathcal{B}_2^{>R}(I, J)|}{L^2} \leq \frac{\Xi g}{RL} \ll Q^{1-\alpha-\lambda} = Q^{\eta-\alpha} = o(1). \quad (75)$$

Adding (74) and (75) proves the theorem. $\square$

Remark 3.11 (A method range, not a barrier). The two strict requirements in (73) have different origins. Burgess suppression needs $\alpha < 4(1 - \eta)$, whereas the exact high-conductor multiplier needs $\alpha > \eta$. Their overlap is equivalent to $\eta < 4/5$. This proves neither the endpoint nor failure beyond it; it is only the range furnished by this particular cutoff argument. Furthermore, (72) concerns an unweighted finite local kernel. It is not a prime-pair asymptotic or a twin-prime lower bound.

4 A shifted-divisor square which removes the one-prime modes

This section isolates an algebraic point which is useful later and easy to misstate. An ordinary Selberg divisor square supported on the same primes that define a rough set becomes constant on that rough set. Moving the divisor primes outside the roughness window avoids the collapse, but creates a separate distribution interface. A tensor product of nonconstant local weights also cannot remove every one-prime Fourier mode. We therefore use a different object: a square built from the shifted divisibility events $p \mid a - \rho$. Its coefficients can be chosen so that its centered part has exactly two-prime Fourier support.

The phrase *shifted-divisor Selberg-form square* below refers only to this quadratic shape. It is not the standard same-window Selberg weight $(\sum_{d|n} \lambda_d)^2$, and it is not a prime-detecting weight. All statements in this section are finite identities on residue groups.

4.1 Why an ordinary same-window divisor square collapses

Let $\mathcal{P}$ be a finite set of primes and put

$$Q = \prod_{p \in \mathcal{P}} p, \quad G = (\mathbb{Z}/Q\mathbb{Z})^\times. \quad (76)$$

No asymptotic restriction on $\mathcal{P}$ is needed in the first three results.

Proposition 4.1 (Same-window collapse). *Let $(\lambda_d)_{d|Q}$ be arbitrary complex coefficients and set*

$$\nu_{\lambda, Q}(n) = \left| \sum_{\substack{d|Q \\ d|n}} \lambda_d \right|^2. \quad (77)$$

Then

$$\nu_{\lambda, Q}(n) = |\lambda_1|^2 \quad \text{whenever } (n, Q) = 1. \quad (78)$$

Consequently, the restriction of $\nu_{\lambda, Q}$ to G is constant and its centered multiplicative Fourier transform is identically zero. The same conclusion holds if the divisor sum is further restricted by $d \leq D$ for any $D \geq 1$.

Proof. If $d \mid Q$, $d \mid n$, and $(n, Q) = 1$, then $d = 1$. Thus only the $d = 1$ summand remains in (77). A cutoff on d does not alter this observation. $\square$

This is an exact support obstruction, not a failure to optimize the coefficients λ_d . In particular, a normalized choice $\lambda_1 = 1$ gives the constant weight one on the Q -rough support and cannot be tuned there to cancel selected nontrivial characters.

There is a legitimate off-window alternative, but it changes the analytic problem. The following elementary identity makes the interface explicit.

Lemma 4.2 (Off-window interface). *Let R be squarefree and define*

$$R_{\text{off}} = \frac{R}{(R, Q)} = \prod_{\substack{\ell \mid R, \ell \text{ prime} \\ \ell \nmid Q}} \ell, \quad (79)$$

For arbitrary coefficients $(\lambda_d)_{d \mid R}$ and every $(n, Q) = 1$,

$$\left| \sum_{\substack{d \mid R \\ d \mid n}} \lambda_d \right|^2 = \left| \sum_{\substack{d \mid R_{\text{off}} \\ d \mid n}} \lambda_d \right|^2. \quad (80)$$

In particular, only primes outside $\mathcal{P}$ can make this ordinary divisor square vary on the Q -rough set.

Moreover, if the right side of (80) is denoted by $\nu_{\text{off}}(n)$, then for every $F : G \rightarrow \mathbb{C}$,

$$\begin{aligned} & \frac{1}{\varphi(Q)R_{\text{off}}} \sum_{\substack{n \bmod QR_{\text{off}} \\ (n, Q) = 1}} F(n \bmod Q) \nu_{\text{off}}(n) \\ &= \left(\frac{1}{\varphi(Q)} \sum_{a \in G} F(a) \right) \left(\frac{1}{R_{\text{off}}} \sum_{x \bmod R_{\text{off}}} \nu_{\text{off}}(x) \right). \end{aligned} \quad (81)$$

Proof. A divisor of n cannot contain a prime which divides Q , proving (80). Since $(Q, R_{\text{off}}) = 1$, the Chinese remainder theorem identifies the summation set in (81) with $G \times \mathbb{Z}/R_{\text{off}}\mathbb{Z}$, on which the two factors of the summand depend on separate coordinates. $\square$

Thus an off-window divisor square has a nontrivial role only after one controls the joint distribution of the observation interval in the Q -coordinate and the R_{off} -coordinate. Over the complete CRT product it is independent of every centered function of the Q -coordinate. On an incomplete interval the difference from (81) is a genuine discrepancy or character-sum problem; it is not supplied by the Selberg-form algebra.

4.2 A tensor-product no-go statement

Write $G \cong \prod_{p \in \mathcal{P}} \mathbb{F}_p^\times$ and use normalized counting measure on every factor. For $f : G \rightarrow \mathbb{C}$ our Fourier convention is

$$\widehat{f}(\chi) = \mathbb{E}_{a \in G} f(a) \overline{\chi(a)}, \quad f(a) = \sum_{\chi \in \widehat{G}} \widehat{f}(\chi) \chi(a). \quad (82)$$

The support of $\chi = \bigotimes_p \chi_p$ is

$$\text{supp}(\chi) = \{p \in \mathcal{P} : \chi_p \neq \mathbf{1}\}. \quad (83)$$

A one-prime mode is a character whose support has cardinality one.

Proposition 4.3 (Tensor-product no-go). *For each $p \in \mathcal{P}$, let $v_p : \mathbb{F}_p^\times \rightarrow \mathbb{R}_{\geq 0}$ have positive mean m_p , and put*

$$V(a) = \prod_{p \in \mathcal{P}} v_p(a_p), \quad m = \mathbb{E}_G V = \prod_{p \in \mathcal{P}} m_p, \quad V^\circ = V/m - 1. \quad (84)$$

If every one-prime Fourier coefficient of V° vanishes, then every v_p is constant. Hence a nonconstant tensor-product local weight cannot annihilate all one-prime modes.

Proof. If χ is nontrivial only at p , Fourier factorization gives

$$\widehat{V^\circ}(\chi) = \frac{\widehat{v}_p(\chi_p)}{m_p}. \quad (85)$$

The assumed vanishing therefore removes every nontrivial Fourier coefficient of every v_p . Fourier inversion on $\mathbb{F}_p^\times$ makes each v_p constant. $\square$

For example, for any residue vector ρ , the raw local avoidance indicator $\prod_p(1 - \mathbf{1}_{a_p=\rho_p})$ is a nonconstant tensor product and retains one-prime modes. Cross-prime terms are therefore necessary for the exact annihilation sought here.

4.3 The shifted-divisor quadratic design

Assume from now on that every $p \in \mathcal{P}$ is at least 5. Fix a unit residue vector $\rho = (\rho_p)_{p \in \mathcal{P}} \in G$ and define

$$X_{p,\rho}(a) = \mathbf{1}_{a_p=\rho_p} = \mathbf{1}_{p|a-\rho}, \quad \theta_p = \mathbb{E}_G X_{p,\rho} = \frac{1}{p-1}, \quad Y_{p,\rho} = X_{p,\rho} - \theta_p. \quad (86)$$

In the divisibility notation, ρ denotes any CRT lift of the residue vector. Set

$$A = \sum_{p \in \mathcal{P}} \frac{1}{p-3}, \quad C = \frac{2}{1+2A}, \quad t_p = C \frac{p-1}{p-3}, \quad (87)$$

and introduce the nonnegative square

$$W_\rho(a) = \left(1 - \sum_{p \in \mathcal{P}} t_p X_{p,\rho}(a) \right)^2. \quad (88)$$

It is a local majorant for the finite residue-avoidance event:

$$W_\rho(a) \geq \prod_{p \in \mathcal{P}} (1 - X_{p,\rho}(a)). \quad (89)$$

Indeed, both sides equal one when every $X_{p,\rho}$ is zero, while otherwise the right side is zero and the left side is a square. This observation is only about avoidance of one residue modulo each p ; it does not majorize or detect the primes.

Theorem 4.4 (Exact pair-only coordinate decomposition). *Let*

$$\mu = \mathbb{E}_G W_\rho = \frac{C^2}{4} + \sum_{p \in \mathcal{P}} t_p^2 \theta_p (1 - \theta_p) > 0 \quad (90)$$

and $U_\rho = W_\rho/\mu - 1$. Then the pointwise identities

$$W_\rho - \mu = 2 \sum_{\substack{p,q \in \mathcal{P} \\ p < q}} t_p t_q Y_{p,\rho} Y_{q,\rho}, \quad (91)$$

$$U_\rho = \frac{2}{\mu} \sum_{p < q} t_p t_q Y_{p,\rho} Y_{q,\rho} \quad (92)$$

hold. In particular,

$$\mathbb{E}_G(U_\rho \mid a_p) = 0 \quad \text{for every } p \in \mathcal{P}, \quad (93)$$

so the constant and all one-prime coordinate components vanish.

More precisely, with the convention (82),

$$\widehat{U}_\rho(\chi) = \begin{cases} \frac{2t_p t_q}{\mu(p-1)(q-1)} \overline{\chi_p(\rho_p) \chi_q(\rho_q)}, & \text{supp}(\chi) = \{p, q\}, \ p < q, \\ 0, & |\text{supp}(\chi)| \neq 2. \end{cases} \quad (94)$$

Thus the Fourier support is exactly on two-prime characters whenever $|\mathcal{P}| \geq 2$.

Proof. Put $m = \sum_p t_p \theta_p$. From (87),

$$m = CA, \quad C = 2(1 - m). \quad (95)$$

For a Bernoulli variable of mean θ_p ,

$$Y_{p,\rho}^2 = \theta_p(1 - \theta_p) + (1 - 2\theta_p)Y_{p,\rho}. \quad (96)$$

Expanding $W_\rho = (1 - m - \sum_p t_p Y_{p,\rho})^2$, and then using (96), gives

$$\begin{aligned} W_\rho &= (1 - m)^2 + \sum_p t_p^2 \theta_p(1 - \theta_p) \\ &\quad + \sum_p t_p [-2(1 - m) + t_p(1 - 2\theta_p)] Y_{p,\rho} + 2 \sum_{p < q} t_p t_q Y_{p,\rho} Y_{q,\rho}. \end{aligned} \quad (97)$$

Now

$$t_p(1 - 2\theta_p) = C \frac{p-1}{p-3} \frac{p-3}{p-1} = C = 2(1 - m), \quad (98)$$

so every linear coefficient vanishes. Also $(1 - m)^2 = C^2/4$, which identifies the constant in (97) with μ and proves (91).

Independence and $\mathbb{E}Y_{p,\rho} = 0$ give (93). Finally, for a local character χ_p ,

$$\widehat{Y_{p,\rho}}(\chi_p) = \begin{cases} 0, & \chi_p = \mathbf{1}, \\ (p-1)^{-1} \overline{\chi_p(\rho_p)}, & \chi_p \neq \mathbf{1}. \end{cases} \quad (99)$$

Taking the product of the two local transforms in (92) proves (94). $\square$

Remark 4.5 (Why this is not the collapsed weight). If the components of ρ come from one integer, then $X_{p,\rho}(a) = \mathbf{1}_{p|a-\rho}$. Thus (88) uses shifted divisors of $a - \rho$, whereas [proposition 4.1](#) concerns divisors of the already Q -rough integer a . This shift is precisely why the new square is not constant on G . Calling it “Selberg-form” records the square of a linear divisor sum; it does not import the prime-detection properties of a sieve.

4.4 An exact certificate at $Q = 385$

Take $\mathcal{P} = \{5, 7, 11\}$, $Q = 385$, and $\rho = 1$. Then

$$A = \frac{7}{8}, \quad C = \frac{8}{11}, \quad (t_5, t_7, t_{11}) = \left(\frac{16}{11}, \frac{12}{11}, \frac{10}{11} \right), \quad \mu = \frac{93}{121}. \quad (100)$$

If $S(a) = \{p : X_{p,1}(a) = 1\}$, the value of $W_1(a)$ and the number of residues in G with a given pattern are as follows.

$S(a)$	multiplicity	$W_1(a)$
$\emptyset$	135	1
$\{5\}$	45	25/121
$\{7\}$	27	1/121
$\{11\}$	15	1/121
$\{5, 7\}$	9	289/121
$\{5, 11\}$	5	225/121
$\{7, 11\}$	3	1
$\{5, 7, 11\}$	1	729/121

The multiplicities sum to $\varphi(385) = 240$, and their weighted average is $93/121$. The pointwise centered identity is

$$W_1 - \frac{93}{121} = \frac{384}{121}Y_{5,1}Y_{7,1} + \frac{320}{121}Y_{5,1}Y_{11,1} + \frac{240}{121}Y_{7,1}Y_{11,1}. \quad (101)$$

Since every character takes the value one at $\rho = 1$, the nonzero Fourier coefficients of $U_1 = W_1/\mu - 1$ are exactly

$$\hat{U}_1(\chi) = \begin{cases} 16/93, & \text{supp}(\chi) = \{5, 7\}, \\ 8/93, & \text{supp}(\chi) = \{5, 11\}, \\ 4/93, & \text{supp}(\chi) = \{7, 11\}. \end{cases} \quad (102)$$

There are respectively 15, 27, and 45 characters of these three types. Parseval therefore gives the exact check

$$\mathbb{E}_G |U_1|^2 = 15 \left(\frac{16}{93}\right)^2 + 27 \left(\frac{8}{93}\right)^2 + 45 \left(\frac{4}{93}\right)^2 = \frac{2096}{2883}. \quad (103)$$

In addition, for every $p \in \{5, 7, 11\}$ and every $x \in \mathbb{F}_p^\times$,

$$\sum_{\substack{a \in G \\ a_p = x}} U_1(a) = 0. \quad (104)$$

Equations (100)–(104) are rational identities. The accompanying exact-arithmetic program and JSON file reproduce the pattern table, the fiber sums, and Parseval mass. They are finite consistency certificates, not numerical evidence for a prime-pair assertion.

4.5 Pairing with the centered local kernel

We finally quantify the gain produced by removing the one-prime modes. For an integer h , let

$$\mathcal{P}_h = \{p \in \mathcal{P} : p \nmid h\}, \quad \kappa_h = \prod_{p \in \mathcal{P}_h} \frac{p-2}{p-1}, \quad (105)$$

and, for $a, b \in G$, define

$$K_h^\circ(a, b) = \kappa_h^{-1} \prod_{p \in \mathcal{P}_h} \mathbf{1}_{a_p b_p \not\equiv -h \pmod{p-1}}. \quad (106)$$

The empty product convention makes $K_h^\circ = 0$ when $\mathcal{P}_h = \emptyset$. Let

$$(T_h f)(a) = \mathbb{E}_{b \in G} K_h^\circ(a, b) f(b). \quad (107)$$

We use the Hermitian convention $\langle f, g \rangle_G = \mathbb{E}_G f \bar{g}$. Since U_ρ , U_σ , and K_h° are real-valued, the bilinear average below equals the indicated inner product:

$$\mathcal{C}_h(\rho, \sigma) = \mathbb{E}_{a, b \in G} U_\rho(a) U_\sigma(b) K_h^\circ(a, b) = \langle U_\rho, T_h U_\sigma \rangle_G. \quad (108)$$

Theorem 4.6 (Pair-only kernel bound). *For all $\rho, \sigma \in G$ and all integers h ,*

$$|\mathcal{C}_h(\rho, \sigma)| \leq \frac{4}{\mu^2} \sum_{\substack{p < q \\ p, q \in \mathcal{P}_h}} t_p^2 t_q^2 \theta_p^2 \theta_q^2 \quad (109)$$

$$\leq 32 \left(\sum_{p \in \mathcal{P}_h} \frac{1}{(p-3)^2} \right)^2. \quad (110)$$

In particular, if

$$\mathcal{P} = \{p : w < p \leq w^2\}, \quad (111)$$

then, as $w \rightarrow \infty$,

$$\sup_{h \in \mathbb{Z}} \sup_{\rho, \sigma \in G} |\mathcal{C}_h(\rho, \sigma)| \ll \frac{1}{(w \log w)^2}. \quad (112)$$

The implied constant is absolute.

Proof. The multiplicative-character expansion of (106) contains only characters which are trivial outside $\mathcal{P}_h$. A character with support $S \neq \emptyset$ has coefficient of magnitude

$$\prod_{p \in S} \frac{1}{p-2}. \quad (113)$$

By (94), only supports $S = \{p, q\}$ can contribute to (108). For a fixed pair $p < q$ there are $(p-2)(q-2)$ such characters. Each kernel coefficient has magnitude $((p-2)(q-2))^{-1}$, while each of the two weight coefficients has magnitude

$$\frac{2t_p t_q \theta_p \theta_q}{\mu}. \quad (114)$$

The character count and the kernel denominator therefore cancel exactly. The triangle inequality, summed over active pairs, proves (109). This calculation also fixes the Fourier normalization: no factor $|G|$ occurs because every average and Fourier coefficient is normalized as in (82).

From (87) and $\theta_p = (p-1)^{-1}$,

$$t_p^2 \theta_p^2 = \frac{C^2}{(p-3)^2}. \quad (115)$$

Also (90) gives $\mu \geq C^2/4$. Hence

$$\begin{aligned} \frac{4}{\mu^2} \sum_{p < q} t_p^2 t_q^2 \theta_p^2 \theta_q^2 &\leq \frac{2C^4}{\mu^2} \left(\sum_p \frac{1}{(p-3)^2} \right)^2 \\ &\leq 32 \left(\sum_p \frac{1}{(p-3)^2} \right)^2, \end{aligned} \quad (116)$$

which proves (110). Finally, the standard upper bound $\pi(x) \ll x/\log x$, followed by partial summation, yields

$$\sum_{w < p \leq w^2} \frac{1}{(p-3)^2} \ll \frac{1}{w \log w}. \quad (117)$$

Substitution in (110) proves (112). $\square$

Remark 4.7 (Scope of the suppression). The saving in (112) is an exact finite Fourier pairing estimate for two deliberately designed residue weights and the centered local kernel. It does not estimate von Mangoldt correlations, does not resolve the sieve parity problem, and does not give a lower bound for any prime-pair count. Its role is narrower: within the local model it removes the entire one-prime obstruction and measures the remaining two-prime contribution without suggesting that the same cancellation holds for primes.

5 Exact finite certificates

The accompanying reference implementation uses only integers and rational numbers. It has three roles: to verify the finite inclusion–exclusion fiber bound, to pin down the shifted-square Fourier normalizations, and to check that the conductor decompositions reconstruct the directly counted finite kernel together with the elementary low/high conductor envelopes. These are exact finite diagnostics, not inputs from which an asymptotic statement is inferred.

Take

$$Q = 5 \cdot 7 \cdot 11 = 385, \quad |G| = 240. \quad (118)$$

The shifted-divisor weight parameters, pattern table, Fourier coefficients, and exact Parseval identity at this modulus are given once in [section 4.4](#). The same rational program verifies those identities. We now record the separate conductor decomposition of the local pair kernel.

For the intervals

$$I = [19, 101], \quad J = [73, 155], \quad (119)$$

both of length 83, the unit counts are $U_I = 50$ and $U_J = 51$. Direct counting at $h = 2$ gives $N_2 = 1439$, $\kappa = 9/16$, and hence

$$\mathcal{B}_2 = \kappa^{-1} N_2 - U_I U_J = \frac{74}{9}. \quad (120)$$

The exact-conductor decomposition is shown in [table 1](#).

Table 1: Exact conductor contributions to the $Q = 385$ reference box.							
q	5	7	11	35	55	77	385
$\mathcal{B}_{2,q}$	2/3	0	-10/9	92/15	82/27	4/9	-128/135
$\mathcal{E}_q$	4/3	24/25	160/81	952/75	10288/729	7316/2025	348044/18225

The fixed contributions sum to $74/9$. Across all 240 unit shifts, the exact mean of $\mathcal{B}_{2\ell}$ is zero and the exact mean square is

$$\frac{1}{240} \sum_{\ell \in G} |\mathcal{B}_{2\ell}|^2 = \sum_{q|385, q>1} \mathcal{E}_q = \frac{65348}{1215}. \quad (121)$$

The tests also cover all seven nonempty local supports, intervals crossing zero and complete periods, translation by Q , exchange of the two factor intervals, inactive primes, and the low/high conductor split.

6 What the theorems do and do not say

The fixed-shift conclusion can be rewritten as an equidistribution statement for one finite local condition. For $(h, Q) = 1$, let

$$N_h(I, J) = \#\{(m, n) \in I \times J : (mn, Q) = 1, (mn + h, Q) = 1\}.$$

Then

$$N_h(I, J) = \kappa(U_I U_J + \mathcal{B}_h(I, J)). \quad (122)$$

Consequently, $\mathcal{B}_2 = o(L^2)$ says that the finitely many congruence exclusions defining Q are equidistributed over the rough factor box at the ambient L^2 scale. It does not say that m , n , or $mn + 2$ is prime.

6.1 Four interfaces visible from the local model

The following list is neither exhaustive nor sufficient for a twin-prime theorem. It records four interfaces for the present route that the local model makes visible; other routes may organize the missing input differently, and additional obstacles may remain.

- G1. Prime-sensitive coefficients.** The interval transforms in this paper have coefficient 1 on $(n, Q) = 1$. Replacing them by Λ , a prime-supported sequence, or a bilinear convolution requires new character estimates. Burgess for the unweighted character sum does not provide those estimates.
- G2. A genuine Type II transfer.** The exact finite kernel supplies a useful conductor multiplier, but a prime-sensitive continuation would need new distribution input for the arithmetic coefficients, for example low-conductor estimates together with high-conductor bilinear control. Well-factorability of a modulus weight does not automatically give either property to the factor coefficients.
- G3. The prescribed-shift and positivity problem.** Averaging shifts diagonalizes the finite phases. A prime-sensitive theorem for $h = 2$ would need a corresponding fixed-shift analysis. To prove infinitely many prime pairs, one would need a positive lower bound in a clearly specified prime-counting range X ; if the target is the Hardy–Littlewood order of magnitude, its expected scale is $X/(\log X)^2$. An ambient $o(L^2)$ error supplies neither statement.
- G4. Parity.** Local divisibility data cannot distinguish integers with an even or odd number of unsieved prime factors. A later argument must add input that bypasses the classical parity obstruction; no such input is present here.

Remark 6.1 (Method ranges, not barriers). The strict ranges $\eta < 1$ for shift means and $\eta < 4/5$ for a fixed local shift are ranges furnished by the present combination of Burgess and the exact high-conductor multiplier. We do not assert either endpoint, optimality, or impossibility beyond them.

6.2 A precise next interface

The next analytic layer can be formulated without a conjectural conclusion. For a structured sequence α_n , define

$$A_{I,\alpha}(\chi) = \sum_{\substack{n \in I \\ (n,Q)=1}} \alpha_n \chi(n).$$

After defining the correspondingly weighted box form, the character expansion and complete-shift diagonalization of [section 2](#) retain the same algebraic form with A_I replaced by $A_{I,\alpha}$. The Parseval energies, high-conductor estimates, and completion error must instead be recomputed with the appropriate weighted norms. Progress toward prime-sensitive applications would therefore require one or both of the following independently checkable statements:

- (a) a Siegel–Walfisz-type estimate, or a zero-density argument implying an appropriate averaged transform bound, at low conductor and uniformly in the intended short intervals;
- (b) a bilinear or dispersion estimate that sums the high-conductor transforms more efficiently than the pointwise multiplier cutoff.

If either statement fails for a proposed weight, that failure is itself a rigorous obstruction result for that route. If both become available, the exact kernel formulas show where they enter, but positivity and parity would still remain separate tasks.

7 Conclusion

The finite CRT kernel has enough spectral structure to turn classical character-sum information into power-short rough-box theorems. Conductor-by-conductor aggregation is essential: low conductors are controlled by the primitive character sum, while high conductors are damped by the exact multiplier. Burgess then yields normalized cancellation on every fixed power-length box after fully-active shift averaging, and up to $\eta < 4/5$ for the fixed local shift $h = 2$.

The shifted-divisor quadratic design gives a complementary finite mechanism. It is non-negative, majorizes the desired residue-avoidance indicator, and removes all one-prime modes exactly by coupling different prime coordinates. The same-window collapse result explains why an ordinary divisor square supported only on divisors of Q cannot accomplish this on the unit group.

These are strict local advances, not prime-pair theorems. The resulting framework does, however, replace a vague analogy by a concrete division of labor: classical conductor estimates at low frequency, exact kernel damping at high frequency, and a non-exhaustive list of prime-sensitive, fixed-shift, positivity, and parity interfaces not controlled here.

Data and code availability

The manuscript source, exact rational certificate, and standard-library Python tests are included in the repository directory for this paper. Running

```
python exact_diagnostics.py --output data/exact-certificate.json
python -m unittest -v test_exact_diagnostics.py
```

inside the `experiments` directory regenerates the data and verifies the identities. No random seed, floating-point Fourier transform, or external data set is used.

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